Separation, Pooling, and Endogenous Timing

A Tullock Contest under Two-sided Asymmetry

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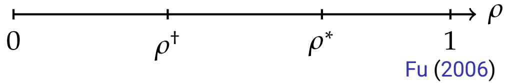
Research Question

- **Fu (2006)**: in a Tullock contest with private information, the informed player prefers to move *second* — revealing type early is costly.
- But this rests on $\rho = 1$: prize values are *perfectly correlated*.
- **Gap**: does the late-mover preference survive at $\rho < 1$, or is it an artifact of full correlation?
- **This paper**: introduce $\rho \in [0, 1]$ — at $\rho = 0$ the signal channel is closed; at $\rho = 1$ Fu's result is recovered as the limiting case.



What This Paper Does

- ***i.i.d.* benchmark** $(\rho = 0)$ — isolates timing from information spillover.
- **Partial correlation** $(0 < \rho < 1)$ as the main model — $\rho = 1$ is the endpoint, not a separate model.
- Characterizes undistorted vs. Riley distorted separation, and their transition as ρ rises.
- Links subgame equilibria to **endogenous timing**: a structural asymmetry between I and U.

Related Literature

Strand	Key reference	Main takeaway
Contest success function	Tullock (1980)	Ratio-form CSF; unique equilibrium at $r = 1$
Exogenous timing	Dixit (1987)	Stronger player overcommits first (top-dog)
Endogenous timing	Baik and Shogren (1992)	Weaker player moves first
Timing framework	Hamilton and Slutsky (1990)	SS / UI / IU timing architecture
Asymmetric info (SS)	Hurley and Shogren (1998)	Type uncertainty suppresses U's effort
Information advantage	Wärneryd (2003)	Informed player can win less

Gap and Contribution

Strand	Key reference	Main takeaway
Signaling in contests	Fu (2006)	$\rho = 1$; late mover avoids revelation cost
Least-cost separation	Riley (1979)	Least-cost separation under binding IC
Timing reversal	Protopappas (2023, 2025)	Early-mover equilibrium under cost asymmetry
Gap	—	*i.i.d.* → fully correlated transition uncharacterized
This paper	—	One ρ spans the spectrum; reopens the signal channel

Model Setup

- Two players: informed (I) and uninformed (U)
- Types: $V_I, V_U \in \{v_H, v_L\}$, $\Pr(V_k = v_H) = q \in (0, 1)$
- **Benchmark assumption**: $V_I \perp V_U$ (*i.i.d.*, $\rho = 0$)

	$V_U = v_H$	$V_U = v_L$
$V_I = v_H$	q^2	$q(1-q)$
$V_I = v_L$	$q(1-q)$	$(1-q)^2$

- Three-stage timing: Stage 0 (choose timing) → Nature draws types → effort subgame
- Three regimes: SS (simultaneous), UI (U moves first), IU (I moves first)

The Closed Signal Channel

Lemma (Conditional Expectation Invariance)

If $V_I \perp V_U$, then

$$\mathbb{E}[V_U \mid x_I] = \mathbb{E}[V_U], \ \forall x_I.$$

→ **Signal channel closed**: beliefs update via Bayes, but U's best response is unchanged — *updating happens, but it doesn't bite*.

Proposition (Unique Separating Equilibrium)

The IU subgame admits a *unique separating equilibrium*, with IC slack

$$\frac{(v_H - v_L)^2}{4\bar{v}} > 0, \qquad \bar{v} \equiv \mathbb{E}[V_k].$$

→ I's lead reveals her type at *no cost*: the IC constraint never binds.

Equilibrium Payoffs across Regimes

Composite coefficients: $\hat{A} \equiv \bar{v}\,\mathbb{E}[V_I^{-1/2}]$, $\hat{B} \equiv \bar{v}\,\mathbb{E}[V_I^{-1}] \geq 1$ (Jensen).

Regime	x_U^*	$\widetilde{\pi}_I$	$\widetilde{\pi}_U$
SS	$\dfrac{\hat{A}^2}{(1+\hat{B})^2}$	$\mathbb{E}\Big[\big(\sqrt{V_I} - \frac{\hat{A}}{1+\hat{B}}\big)^2\Big]$	$\dfrac{\hat{A}^2\hat{B}}{(1+\hat{B})^2}$
UI	$\dfrac{\hat{A}^2}{4}$	$\mathbb{E}\Big[\big(\sqrt{V_I} - \frac{\hat{A}}{2}\big)^2\Big]$	$\dfrac{\hat{A}^2}{4}$
IU (sep.)	$\dfrac{v(2\bar{v}-v)}{4\bar{v}}$	$\dfrac{\mathbb{E}[V_I^2]}{4\bar{v}}$	$\dfrac{\mathbb{E}[(2\bar{v}-V_I)^2]}{4\bar{v}}$

→ **First-mover advantage for** U: $\widetilde{\pi}_U^{UI} - \widetilde{\pi}_U^{SS} = \dfrac{\hat{A}^2(1-\hat{B})^2}{4(1+\hat{B})^2} \geq 0.$

Stage-0 Timing Game

Four unilateral deviation gaps:

$$\Delta_I^{UI} \equiv \tilde{\pi}_I^{UI} - \tilde{\pi}_I^{SS}, \quad \Delta_I^{IU} \equiv \tilde{\pi}_I^{IU} - \tilde{\pi}_I^{SS}, \quad \Delta_U^{UI} \equiv \tilde{\pi}_U^{UI} - \tilde{\pi}_U^{SS}, \quad \Delta_U^{SS} \equiv \tilde{\pi}_U^{SS} - \tilde{\pi}_U^{IU}$$

Proposition (Stage-0 Pure-strategy Equilibria)

Pure-strategy Nash equilibria are fully characterized by the signs of the four gaps:

- UI iff $\Delta_I^{UI} \geq 0$
- SS iff $\Delta_I^{UI} \leq 0$ and $\Delta_U^{SS} \geq 0$
- IU iff $\Delta_I^{IU} \geq 0$ and $\Delta_U^{SS} \leq 0$

→ **Takeaway**: timing is driven purely by gap signs — no signaling distortion.
→ Any effect at $\rho > 0$ must come from the reopened channel — not from asymmetric information per se.

Correlated Type Structure

- Pearson correlation: $\rho \equiv \mathrm{corr}(H_I, H_U)$, where $H_k = \mathbf{1}\{V_k = v_H\}, \rho \in [0,1]$
- Marginals fixed at $(q, 1-q)$ — ρ is the single free parameter of the joint distribution

	$V_U = v_H$	$V_U = v_L$
$V_I = v_H$	$q^2 + \rho q(1-q)$	$q(1-q)(1-\rho)$
$V_I = v_L$	$q(1-q)(1-\rho)$	$(1-q)^2 + \rho q(1-q)$

ρ	Structure	Key property
$\rho = 0$	*i.i.d.* benchmark	signal has no spillover
$0 < \rho < 1$	partial correlation	belief enters U's best response
$\rho = 1$	fully correlated	Fu (2006) endpoint

Posterior Expected Valuation

Conditional probabilities of U's high state, given I's type:

$$\theta_H = q + \rho(1-q), \qquad \theta_L = q(1-\rho).$$

Lemma (Posterior Expected Valuation)

Given posterior belief $\mu = \Pr(V_I = v_H \mid x_I)$,

$$\widetilde{V}_U(\mu) = v_L + [q + \rho(\mu - q)]\Delta V, \qquad \frac{d\widetilde{V}_U(\mu)}{d\mu} = \rho\,\Delta V > 0.$$

Posterior μ	Interpretation	$\widetilde{V}_U(\mu)$
$\mu = 1$	I is high type	$v_L + [q + \rho(1-q)]\Delta V$
$\mu = q$	beliefs unchanged	$v_L + q\,\Delta V = \mathbb{E}[V_U]$
$\mu = 0$	I is low type	$v_L + q(1-\rho)\Delta V$

→ **When $\rho > 0$, I's type is a signal about U's own prize — the channel is live.**

SS and UI under Partial Correlation

Composite coefficients — replace $\bar{v}$ with $\widetilde{V}_U(1), \widetilde{V}_U(0)$:

$$\tilde{A} \equiv q\,\frac{\widetilde{V}_U(1)}{\sqrt{v_H}} + (1-q)\,\frac{\widetilde{V}_U(0)}{\sqrt{v_L}}, \qquad \tilde{B} \equiv q\,\frac{\widetilde{V}_U(1)}{v_H} + (1-q)\,\frac{\widetilde{V}_U(0)}{v_L}.$$

- Equilibrium structure identical in form to the benchmark: $\hat{A}, \hat{B} \to \tilde{A}, \tilde{B}$ (at $\rho = 0$: $\tilde{A} = \hat{A}, \tilde{B} = \hat{B}$)

Proposition (First-mover Advantage Robust)

$\widetilde{\pi}_U^{UI} \geq \widetilde{\pi}_U^{SS}$ for all $\rho \in [0, 1)$.

- **Monotone coefficients**: $\tilde{A}, \tilde{B}$ strictly decreasing in ρ, with $\tilde{B} \to 1$ as $\rho \to 1$ — rising ρ erases the Jensen excess burden $\hat{B} > 1$

When Does Separation Hold?

Undistorted separating candidate in the IU subgame, and its IC condition:

$$x_H = \frac{v_H^2}{4\widetilde{V}_U(1)}, \quad x_L = \frac{v_L^2}{4\widetilde{V}_U(0)}; \qquad \frac{\widetilde{V}_U(1)}{\widetilde{V}_U(0)} \le R \equiv \frac{v_H^2}{v_L(2v_H - v_L)}.$$

Proposition (Separation Threshold)

Undistorted separation holds iff $\rho \le \rho^*$, where

$$\rho^* = \frac{(R-1)\,\widetilde{V}_U(q)}{(1-q+Rq)\,\Delta V}.$$

- At $\rho = 0$: strictly satisfied; at $\rho = 1$: violated
- IC slack $S(\rho) \equiv R - \widetilde{V}_U(1)/\widetilde{V}_U(0)$ decreases monotonically and hits 0 exactly at ρ^*

Riley Distortion

When $\rho > \rho^*$, define

$$\alpha \equiv 1 - \frac{\widetilde{V}_U(0)}{\widetilde{V}_U(1)} \in (0,1), \qquad z_H^- \equiv \frac{v_H}{2\sqrt{\widetilde{V}_U(0)}}(1 - \sqrt{\alpha}).$$

- High type keeps undistorted effort x_H; low type distorts *down* to $x_L^R = (z_H^-)^2$

Corollary (Two-region Structure)

- $\rho \leq \rho^*$: undistorted separation — no signaling cost
- $\rho > \rho^*$: Riley distorted separation — separation cost borne by the low type alone

$\rightarrow$ High type's mimicry gain grows with ρ; the low type must undercut to deter it.

Stage-0 Payoff Matrix

Key point: at stage 0, types are unrealized — players compare *ex ante* expected payoffs.

	U: Early	U: Late
I: Early	$(\tilde{\pi}_I^{SS}, \tilde{\pi}_U^{SS})$	$(\tilde{\pi}_I^{IU}, \tilde{\pi}_U^{IU})$
I: Late	$(\tilde{\pi}_I^{UI}, \tilde{\pi}_U^{UI})$	$(\tilde{\pi}_I^{SS}, \tilde{\pi}_U^{SS})$

Proposition (Stage-0 Pure-strategy Equilibria)

Same sign characterization as the benchmark $(\Delta_I^{UI}, \Delta_I^{IU}, \Delta_U^{UI}, \Delta_U^{SS})$ — but now *every entry depends on* ρ.

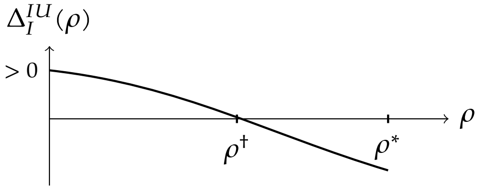
→ The question becomes: **how do the gap signs move with ρ?**

The Structural Asymmetry

- U**'s side**: $\Delta_U^{UI} \geq 0$ for all $\rho < 1$ — U always wants to lead against a late-moving I (robust)
- I**'s side**: $\widetilde{\pi}_I^{IU}$ strictly decreasing in ρ — revelation cost rises with correlation

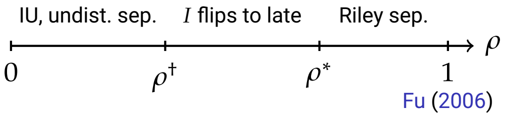
Proposition (Timing Reversal)

If $\Delta_I^{IU}(0) > 0$ and $\Delta_I^{IU}(\rho^*) < 0$, there exists $\rho^\dagger \in (0, \rho^*)$ where I's timing preference flips from early to late.



→ **The reversal is driven entirely by** I**, not** U**.**

Summary — The Correlation Spectrum



ρ range	IU equilibrium type	I's timing preference
$[0, \rho^{\dagger})$	undistorted separation	lead (IU)
$(\rho^{\dagger}, \rho^{*}]$	undistorted separation	wait
$(\rho^{*}, 1)$	Riley distorted separation	wait
$\rho = 1$	Fu (2006) endpoint	wait (late mover)

→ **Bottom line**: Fu (2006)'s late-mover result *requires* correlation — it is the right endpoint, not a general principle.

Welfare Framework

Social welfare = allocative value minus effort dissipation: $\widetilde{W} = \widetilde{G} - \widetilde{T}$, where $\widetilde{G} = \mathbb{E}[p_I V_I + p_U V_U]$ and $\widetilde{T} = \mathbb{E}[x_I + x_U]$.

Equilibrium type	Expected dissipation $\widetilde{T}$	Sensitivity to ρ
SS	$\dfrac{\tilde{A}}{1+\tilde{B}}\,\mathbb{E}[\sqrt{V_I}]$	varies via $\tilde{A}$, $\tilde{B}$
UI	$\dfrac{\tilde{A}}{2}\,\mathbb{E}[\sqrt{V_I}]$	varies via $\tilde{A}$
IU undistorted sep.	$\dfrac{\mathbb{E}[V_I]}{2}$	**immune to ρ**
IU Riley distorted	$\dfrac{v_H}{2}[1-(1-q)\sqrt{\alpha}]$	varies via α
IU pooling	$\sqrt{x_P\,\bar{v}}$	unaffected by ρ

→ **IU undistorted separation's dissipation is insulated from ρ** — the key structural difference.

Three Effects of ρ on Welfare

- **Allocative effect**: $\rho \uparrow \rightarrow \widetilde{V}_U(1)$ and $\widetilde{V}_U(0)$ diverge → type-specific competition in IU matches true values better → allocation improves
- **Dissipation effect**: $\rho \uparrow \rightarrow$ SS dissipation keeps rising, while IU undistorted dissipation stays fixed at $\mathbb{E}[V_I]/2$ → IU's cost advantage widens
- **Threshold effect**: ρ crosses ρ^* → Riley distortion begins → low-type allocation deteriorates — a *discontinuous* welfare drop

$W(\rho)$ is generally non-monotone

- Increasing for $\rho \le \min(\rho^\dagger, \rho^*)$: dissipation is insulated, only the allocative effect operates
- Once the equilibrium shifts to SS at $\rho^\dagger$, the increase stops
- Riley distortion beyond ρ^* further reduces welfare

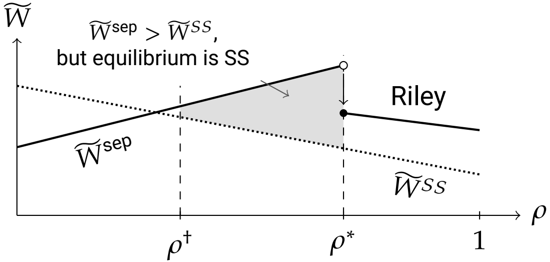
Equilibrium vs. Social Optimum

- **Direction 1** (ρ large): equilibrium is SS, yet $\widetilde{W}^{\text{sep}} > \widetilde{W}^{SS}$ — *I's retreat sustains the high-dissipation regime*
- **Direction 2** (ρ small): equilibrium is IU, yet $\widetilde{W}^{SS} > \widetilde{W}^{IU}$ — *Riley distortion degrades the low-type allocation*

Root cause

I does not internalize U's payoff; private timing incentives ignore the global dissipation–allocation trade-off. Efficiency of equilibrium timing is *not* a general result — it depends on the parameter configuration.

Welfare across Regimes



- $\widetilde{W}^{\text{sep}}$ rises in ρ (fixed dissipation, improving allocation); $\widetilde{W}^{SS}$ declines (rising dissipation)
- Shaded region: socially IU is better, but I has already abandoned the lead at $\rho^{\dagger}$
- Discontinuous welfare drop at ρ^{*} (Riley distortion)

Conclusion

1. $\rho = 0$: unique undistorted separation – I leads at *no revelation cost*.
2. $\rho > 0$ reopens the signal–belief–response channel: beliefs enter U's best response.
3. **Separation threshold**: undistorted iff $\rho \leq \rho^*$; beyond, Riley distortion – cost borne by the low type alone.
4. **Timing asymmetry**: U's lead preference is robust; I flips to late at $\rho^\dagger < \rho^*$.
5. Fu (2006) is the $\rho = 1$ endpoint – not a general principle.
6. **Welfare**: equilibrium timing departs from the social optimum; $W(\rho)$ is non-monotone.

Future Directions

1. **Continuous type space** — beyond the closed-form threshold: richer distortion patterns.
2. **Negative correlation** ($\rho < 0$) — reverse information spillover.
3. N**-player contests** — timing symmetry-breaking; the role of correlation remains open.